

ETF rotation backtest

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A rule-based ETF portfolio tested in QuantConnect from 1 March 2010 to 10 July 2026.

Research result, not a validated live strategy

The portfolio grew from \$100,000 to \$376,259 after modeled costs. This is a full-sample backtest used during strategy development. Statistical evidence of an edge has not been established, and the strategy underperformed the exported SPY benchmark in cumulative return.

Results at a glance

Metric	Result	How to read it
Starting / ending equity	\$100,000 / \$376,259.19	Simulated portfolio value, including open holdings.
Total net return	276.259%	Cumulative growth after modeled trading costs.
Net equity gain	\$276,259.19	Ending equity minus starting equity.
Annualized return (CAGR)	8.430%	Equivalent compounded yearly growth over this sample.
Maximum drawdown	17.600%	Largest peak-to-trough equity loss, as reported by QC.
Sharpe / Sortino	0.458 / 0.474	QC risk-adjusted ratios; Sortino focuses on downside variation.
Probabilistic Sharpe ratio	0.334%	QC confidence relative to its reference Sharpe, not a forecast.
Annual volatility	9.3%	QC annual standard deviation, exported as 0.093.
Total fees	\$8,831.83	Modeled transaction fees over the full run.
Portfolio turnover	7.60%	QC headline statistic; not the custom annual Turn/yr log.
Win / loss rate	64% / 36%	QC portfolio-statistics definition; not a guarantee of profitability.
Average win / loss	0.53% / -0.56%	QC average winning and losing trade returns.
Profit-loss ratio	0.94	Average win divided by the magnitude of average loss.
Orders	1,966	1,965 filled; one invalid order in the export.
Beta / alpha	0.35 / 0.012	QC benchmark sensitivity and estimated regression intercept.
Information ratio	-0.381	Negative benchmark-relative risk-adjusted result.

Values are copied from the QC statistics object unless explicitly labeled as derived. Account currency: USD. Backtest name: Casual Orange Owl. Engine: LEAN v2.5.0.0.18171.

I. Strategy rules

The strategy rotates among 12 ETFs using trend, momentum and market-stress signals. BIL is a separate Treasury-bill ETF used for residual allocation. It normally chooses at most three assets rather than owning the whole universe.

Rotation universe	SPY, QQQ, IWM, EFA, EEM, XLK, XLF, XLE, TLT, GLD, VNQ, HYG
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Reserve allocation	BIL; TQQQ is used only by a separate simulated comparison, not this portfolio.
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Stress signals	VIX / VIX3M term-structure ratio and the 20-day VIX z-score.
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Mode	Trigger and portfolio action
Extreme	Ratio > 1.20 AND VIX z-score > 2.0. At least 50% goes to BIL; up to 50% goes to positive-momentum TLT/GLD.
Reversal	Ratio > 1.05 OR (z-score > 1.5 AND ratio > 1.0). Select up to three ETFs above their 200-day average with positive 3-month returns and the largest volatility-normalized 5-day declines.
Momentum	No stress trigger, SPY above its 200-day average. Rank positive scores above the 100-day average: 40% 1-month + 40% 3-month + 20% 6-month returns, divided by annualized 20-day volatility.
Bear	No stress trigger, SPY not above its 200-day average. Use the momentum rules at 50% gross allocation; the balance goes to BIL.

Modes are evaluated in the order above. Eligible picks are weighted by inverse 20-day volatility, with a 40% cap per asset. Capped weight is redistributed; unallocatable weight goes to BIL. At most two of SPY, QQQ and XLK may be held. This is a narrow concentration rule, not a complete correlation model.

Momentum holdings can remain within the top five ranks. A replacement must improve the score by at least 10% and 0.02 score points. Weight changes below 2% are skipped, except exits. The weighting target is scaled to 99.5%, leaving a small cash buffer.

2. Implementation and timing

The reviewed source uses adjusted daily bars, 420 calendar days of warm-up, an Interactive Brokers margin fee model and ETF-specific constant slippage (1-3 basis points per side for the traded ETFs). These are simulated costs, not live invoices.

Signals use the daily close and submit market-on-open orders for the next session. Rebalancing is decided after the first trading session of each week. Entry into Extreme can happen daily; exit stays weekly. This separates signal formation from intended execution, but does not by itself prove that every data timestamp and fill model is free of look-ahead bias.

3. Growth and loss path

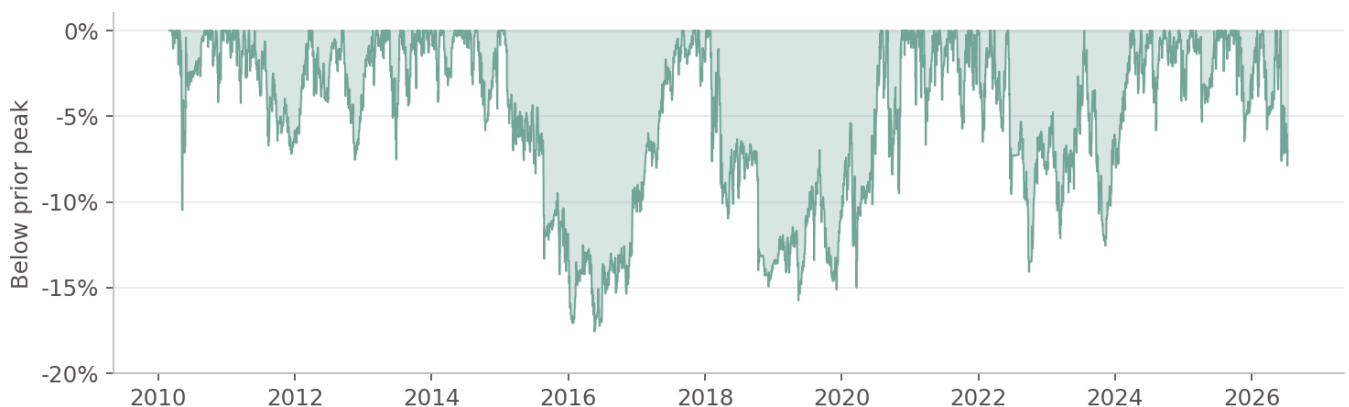
Equity curve and SPY comparison



Strategy equity uses the closing value in each QC equity-chart record. The QC benchmark price series is rebased to the same \$100,000 start. Both use the exported time stamps. A higher line means more cumulative wealth, not necessarily lower risk.

The strategy ends at \$376,259.19; the rebased QC SPY benchmark ends at approximately \$913,748. The strategy therefore did not beat this benchmark in cumulative return. This is the platform benchmark series, not the code's separate fee-bearing BH_SPY simulator. Different cost treatment limits a precise like-for-like comparison.

Drawdown from the previous equity peak

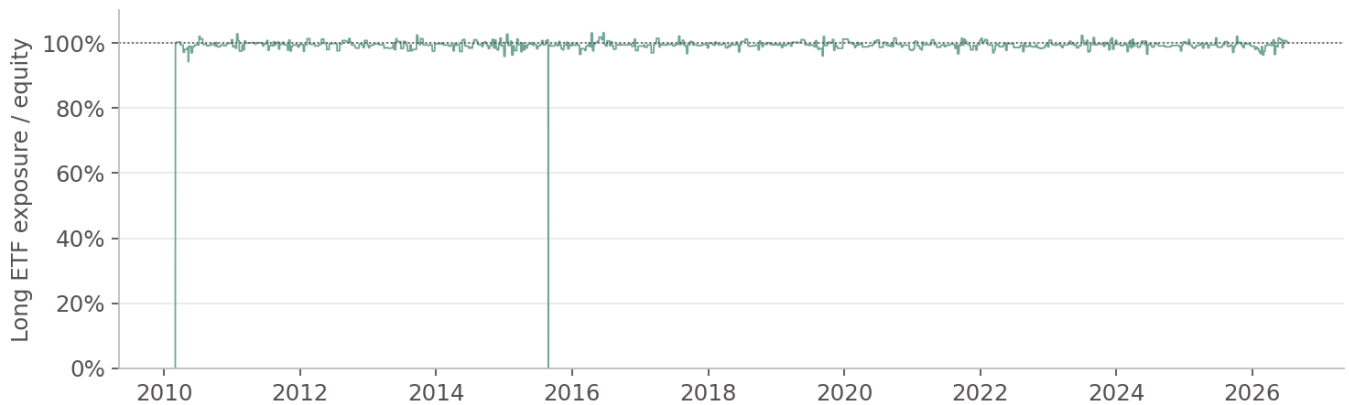


Zero means a new high. -10% means the portfolio is 10% below its prior high. The chart minimum is -17.56%; QC rounds the headline maximum drawdown to 17.600%. Ending drawdown is -7.02%.

Drawdown measures the loss an investor would need to withstand without selling. The long stretches below zero matter as well as the deepest fall: positive final return can coexist with years of uneven progress.

4. Exposure and trading activity

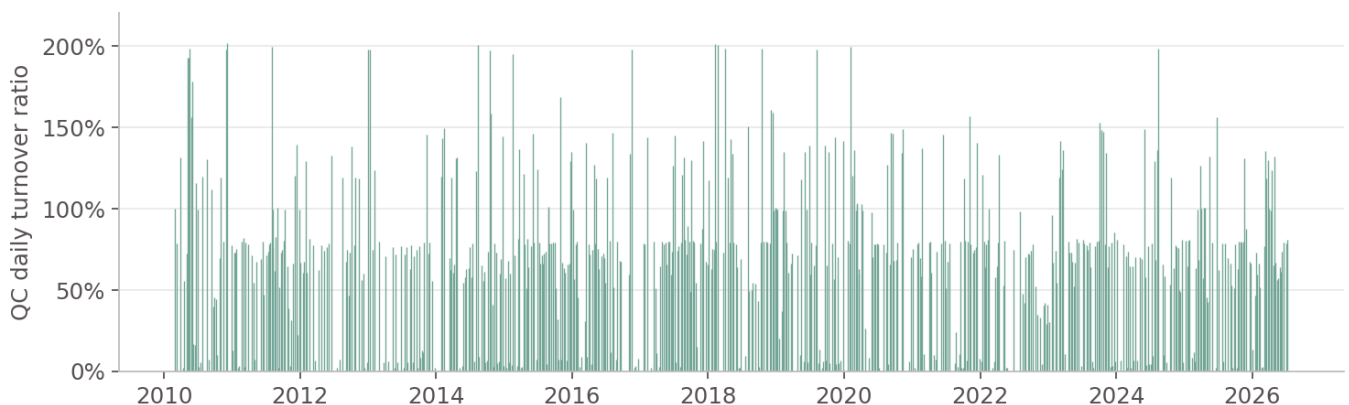
Total long ETF exposure



QC Equity - Long Ratio, multiplied by 100. This includes BIL, bond ETFs and gold as well as stock ETFs. Near 100% does not mean 100% stock-market exposure. All exported short-exposure series are zero.

The maximum sampled long ratio is 103.1%, despite allocation targets below 100%. Target weights are not the same as executed exposure: prices, order timing and the invalid order can cause deviations. The source allows 2x security leverage for buying-power handling; it does not target a 2x portfolio. Actual drift and same-open sell/buy sequencing need review.

Daily portfolio turnover



The exported QC ratio is shown in percent, not as cumulative turnover. Most dates have little or no activity, while replacement dates have large spikes. The largest observation is 201.95%. Replacing a whole portfolio can involve both sells and buys.

QC reports headline turnover of 7.60%. That matches the exported daily series average after conversion to percent (including its calendar-day samples). It must not be presented as the custom annual Turn/yr measure. The code separately logs gross buy-plus-sell notional divided by average equity and years; that log is not included in this JSON.

5. Performance consistency

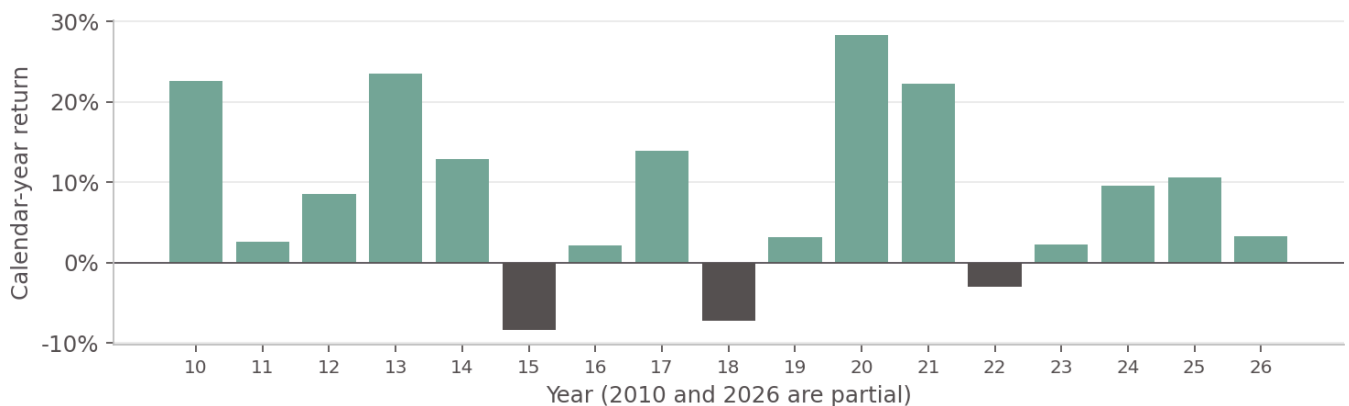
Trailing 12-month return



Copied from the QC M12 rolling-window totalNetProfit values, converted to percent. The final partial July window is excluded; the last complete month is June 2026. Early windows can contain less than 12 months of live strategy history.

Above zero means the measured window gained money; below zero means it lost money. Variation across windows shows why the full-period CAGR alone is not enough. Overlapping windows are not independent experiments.

Calendar-year returns



Derived from daily closing equity, using each year's last exported observation divided by the prior year's last observation. 2010 starts on 1 March and 2026 ends on 10 July, so those bars are partial years, not directly comparable to full years.

These bars separate good and weak periods rather than hiding them inside one total return. They do not attribute gains to individual regimes: that requires holdings and executed-exposure attribution, which has not been completed.

6. Validation still required

This report documents a completed backtest and reviewed rules. It does not claim completed out-of-sample validation, proven alpha or expected future returns.

1. **Development comparison:** run the original and patched versions on March 2010-December 2023 with identical cost and execution assumptions. Compare CAGR, excess Sharpe, drawdown, turnover and regime behavior.
2. **Threshold sensitivity:** run the supplied baseline plus ten one-at-a-time cases. Test reversal ratios 1.03/1.05/1.07, extreme ratios 1.15/1.20/1.25, reversal z-scores 1.25/1.50/1.75, extreme z-scores 1.75/2.00/2.25 and keep-rank 4/5/6. Look for stable neighborhoods rather than selecting the highest-return case.
3. **Freeze, then holdout:** fix the rules before evaluating January 2024 onward. That period has already appeared in the full-sample run, so it is a holdout diagnostic, not genuinely untouched evidence. Future unseen data or paper trading is needed.
4. **Ablations and stress tests:** disable the cluster cap and replacement floor separately, vary sleeve choices one at a time, and test higher fees/slippage, delayed fills and crisis periods. Report unsuccessful variants too.
5. **Execution audit:** resolve the invalid order, confirm data availability and timestamps, and reconcile executed holdings against intended allocations before paper or live use.

Limitations and accounting notes

- **Selection bias:** thresholds and design were developed using this sample. The export's outOfSampleDays metadata does not establish an independent validation experiment.
- **Low PSR:** 0.334% is QC's reported probability relative to its reference Sharpe. It is not the probability of a future profit or a universal probability that the strategy works. The precise reference is not specified in this export; statistical superiority is not established.
- **Execution failure:** order 626, a BIL purchase submitted on 24 August 2015, was invalid due to insufficient buying power. The run completed, but the realized portfolio may differ from its intended target.
- **Simulation limits:** fixed slippage and modeled fees cannot reproduce all liquidity, spread, market-impact or next-open conditions. TLT and GLD are not guaranteed crisis protection. If VIX inputs are unavailable, the code falls back to non-stress defaults; verify coverage.
- **Metric definitions:** headline Sharpe/Sortino are QC statistics, not the source's separate BIL-adjusted custom log metrics. The QC portfolio win rate is 64%; closed-trade statistics separately report 835 winners out of 1,360 trades (61.40%). Do not mix these definitions.
- **Profit reconciliation:** runtime Net Profit is \$277,519.58 and Unrealized is -\$1,281.94. Neither alone equals the \$276,259.19 net equity gain. Fees total \$8,831.83. This report uses equity change and return for the headline, not the runtime profit field.

Sources and reproducibility

Primary evidence: supplied QuantConnect JSON export, completed 8 October 2026, and reviewed patched main.py. No independent backtest rerun was performed for this report. Charts are re-rendered from exported observations without smoothing. Preserve the JSON, exact source, settings and logs to reproduce the run.

Metric references: [QuantConnect backtest report](#); [QuantConnect PSR explanation](#). Reviewed 8 October 2026.

Historical simulation only. No live track record or forward-return claim.